

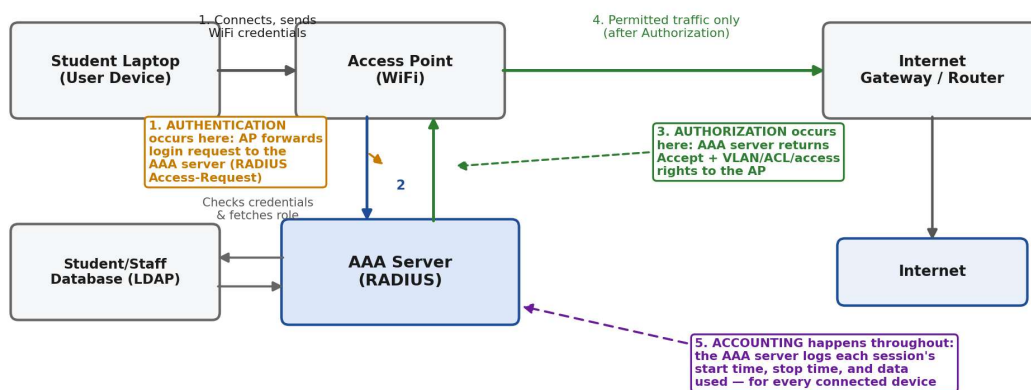
AAA (Authentication, Authorization, Accounting)

Task 1 —Create a table comparing Authentication, Authorization, and Accounting (AAA) by listing at least two real-world examples of each function from popular apps like Instagram, Zomato, or Paytm.

AAA Function	Example 1	Example 2
Authentication	Instagram: logging in with a username and password, or unlocking the app with Face ID/fingerprint.	Paytm: entering a mobile number and OTP to log in, or confirming a payment with the UPI PIN.
Authorization	Instagram: only the admin of a business account can view Insights (content is performing and who are the audience) /ad manager; a regular follower cannot.	Zomato: a verified restaurant partner can update their menu and view incoming orders; a regular customer account cannot.
Accounting	Zomato: every order is logged with time placed, items, amount, and delivery status for tracking and billing.	Paytm: every transaction is recorded in the passbook with timestamp, amount, and parties involved, for audit and dispute resolution.

Task 2 — Draw a simple network diagram showing how AAA is used to secure access to a WiFi network in a college campus, labeling where authentication, authorization, and accounting take place. (Include devices like user laptops, access points, AAA server, and the internet gateway.)

AAA Securing Campus WiFi Access



Task 3 — Explain the difference between centralized and distributed AAA deployment models by describing how each would work in a scenario where multiple Flipkart warehouses need secure network access.

Centralized model: Flipkart sets up one central AAA server, likely at a main data centre or head office, and every warehouse across the country — sends its login requests to this single server over the company's WAN. When a warehouse worker logs into a handheld scanner or a warehouse management terminal, the local network equipment forwards that request all the way to the central server, which checks the credentials, decides what the worker is allowed to access (e.g. inventory scanning but not shipment routing), and logs the session. This keeps policy and logs in one place, but every warehouse depends on connectivity back to that central server to let anyone log in.

Distributed model: instead, each warehouse (or each regional cluster of warehouses, e.g. all North-India warehouses) runs its own local AAA server. A worker logging in at the Bengaluru warehouse authenticates against the Bengaluru server, not one all the way across the country. The local servers periodically sync policy and user lists with each other or with a central management layer, so a worker with company-wide

access can still log in at any warehouse, but the actual authentication happens locally and doesn't depend on a single distant server being reachable.

For Flipkart's scale, a distributed setup is generally more practical: a temporary WAN outage at one warehouse won't lock workers out everywhere else, and login is faster since it's handled by a nearby server rather than one that might be hundreds of kilometres away.

Task 4 — Pick any one AAA deployment model and list three advantages and two disadvantages of using it for a large-scale app like BookMyShow.

Model chosen: Distributed AAA

Advantages:

1. Resilience — if one regional server or data centre goes down (say, during a high-traffic movie or concert ticket release), logins in other regions are unaffected, so BookMyShow doesn't face a nationwide outage from one point of failure.
2. Lower latency — users logging in are authenticated by a nearby server rather than one on the other side of the country, which keeps the app feeling fast, especially important in the seconds-matter rush of a ticket sale.
3. Scalability — as BookMyShow expands into more cities or adds more concurrent users, load can be spread across multiple regional AAA servers instead of overloading one central system.

Disadvantages:

1. Consistency challenges — keeping user accounts, permissions, and security policies synced across many distributed servers is more complex, and a sync delay could mean a policy change (like blocking a compromised account) doesn't take effect everywhere instantly.
2. Higher management overhead — more individual servers means more systems to patch, monitor, and secure, which increases both operational cost and the number of potential attack surfaces compared to managing one central system.

Task 5 — Use ChatGPT or another AI tool to generate a sample AAA policy for a Zomato-like app, specifying what actions are authenticated, which roles are authorized for each action, and what accounting logs should be kept. Paste the generated policy and highlight one thing you would change to make it more secure.

AAA Policy – Food Delivery App ("QuickBite")

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1. AUTHENTICATION (required before any action below)

- Customers: mobile number + OTP, or saved-session token on trusted device.
- Delivery Partners: mobile number + OTP + device ID binding.
- Restaurant Partners: email/username + password, with OTP for first login on a new device.
- Internal Staff/Admins: company SSO login + mandatory MFA (Multi-Factor Authentication).

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2. AUTHORIZATION (role-based access)

- Customer: browse restaurants, place orders, make payments, rate orders.
- Delivery Partner: view assigned orders, update delivery status, view own earnings. Cannot view other partners' data or restaurant financials.
- Restaurant Partner: update own menu/prices, view own incoming orders and payouts. Cannot view other restaurants' data or customer payment details.
- Admin: full access to dashboards, refunds, and account management,

restricted to company network/VPN and MFA-verified sessions.

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3. ACCOUNTING (logs to keep)

- Every login attempt (success/failure) with timestamp, account, device ID.
- Every order placed: items, amount, payment method, timestamps.
- Every payment transaction and refund, with amount and reason code.
- Every admin action (refund issued, account suspended, menu edited by staff), logged with the admin's ID for audit purposes.

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